AI-Powered Healthcare Data Insights and Dashboarding

The given excel data set contains 3 sheets, customer name, medical examination and hospitalisation details.

Customer ID	name
Id1	Hawks, Ms. Kelly
Id2	Lehner, Mr. Matthew D
Id3	Lu, Mr. Phil
Id4	Osborne, Ms. Kelsey
Id5	Kadala, Ms. Kristyn
Id6	Baker, Mr. Russell B.
Id7	Macpherson, Mr. Scott
Id8	Hallman, Mr. Stephen
Id9	Moran, Mr. Patrick R.
Id10	Benner, Ms. Brooke N.
Id11	Fierro Vargas, Ms. Paola Andrea
Id12	Franz, Mr. David
Id13	Foster, Mr. Wade
Id14	Tenorio, Mr. Franklin
Id15	Rios, Ms. Leilani M.
Id16	Viau-Dupuis, Mr. Philippe
Id17	Cronin, Ms. Jennifer A.
Id18	Noordstar, Ms. Christina M.

Customer Names | Medical Examinations | Hospitalisation Details

Customer ID	BMI	HBA1C	Heart Issues	Any Transplants	Cancer history	NumberOfMajorSurgeries	smoker
Id1	47.41	7.47	No	No	No	No major surgery	yes
Id2	30.36	5.77	No	No	No	No major surgery	yes
Id3	34.485	11.87	yes	No	No	2	yes
Id4	38.095	6.05	No	No	No	No major surgery	yes
Id5	35.53	5.45	No	No	No	No major surgery	yes
Id6	32.8	6.59	No	No	No	No major surgery	yes
Id7	36.4	6.07	No	No	No	No major surgery	yes
Id8	36.96	7.93	No	No	No	3	yes
Id9	41.14	9.58	yes	No	Yes	1	yes
Id10	38.06	10.79	No	No	No	No major surgery	yes
Id11	37.7	5.96	yes	No	No	2	yes
Id12	42.13	11.9	No	No	No	No major surgery	yes
Id13	40.92	8.41	No	No	No	No major surgery	yes
Id14	40.565	7.02	No	No	No	No major surgery	yes
Id15	36.385	7.59	yes	No	No	2	yes

Customer ID	year	month	date	children	charges	Hospital tier	City tier	State ID
Id2335	1992	Jul	9	0	563.84	tier - 2	tier - 3	R1013
Id2334	1992	Nov	30	0	570.62	tier - 2	tier - 1	R1013
Id2333	1993	Jun	30	0	600	tier - 2	tier - 1	R1013
Id2332	1992	Sep	13	0	604.54	tier - 3	tier - 3	R1013
Id2331	1998	Jul	27	0	637.26	tier - 3	tier - 3	R1013
Id2330	2001	Nov	20	0	646.14	tier - 3	tier - 3	R1012
Id2329	1993	Jun	1	0	650	tier - 3	tier - 3	R1013
Id2328	1995	Jul	4	0	650	tier - 3	tier - 3	R1013
Id2327	2002	Nov	29	0	668	tier - 3	tier - 2	R1012
Id2326	1997	Nov	9	0	670	tier - 3	tier - 3	R1013
Id2325	2001	Sep	12	0	687.54	tier - 3	tier - 2	R1013
Id2324	1999	Dec	26	0	700	?	tier - 3	R1013
Id2323	1999	Dec	14	0	722.99	tier - 3	tier - 1	R1013
Id2322	2002	?	19	0	750	tier - 3	tier - 1	R1012
Id2321	1993	Aug	9	0	760	tier - 3	tier - 1	R1013
Id2320	1996	Oct	22	0	760	tier - 3	tier - 3	R1013

Data Cleaning:

Remove duplicate

In the first step of data cleaning, count of Customer ID is checked in all three data sheets. The customer names sheet and medical examination sheet have the same Customer Ids. But the hospitalisation details data set contains 8 extra Ids.

Clipboard | Font | Alignment | Number | Styles | Cells

Imputed customer id

Customer	Imputed customer	year	month	date	children	charges	Hospital tier	City tier	State ID
?	?	2004	Nov	6	0	1137.01	tier - 3	tier - 1	R1013
?	?	1999	Jun	9	1	2775.19	tier - 2	tier - 1	R1012
?	?	1985	Dec	20	2	6203.9	tier - 1	tier - 2	R1012
id2444	Not found	1987	Nov	27	2	20984.09	tier - 2	tier - 2	R1015
id3444	Not found	2004	Nov	1	2	34303.17	tier - 1	tier - 3	R1013
?	?	2000	Oct	13	0	35585.58	tier - 1	tier - 2	R1011
?	?	1992	Oct	6	0	36837.47	tier - 1	tier - 2	R1011
?	?	1991	Nov	22	2	38711	tier - 1	tier - 3	R1011

Then the unknown Customer Ids were checked with the "Charges" column. Except the Customer ID all the other data were same in 5 Ids. So we considered that 5 unknown Ids are duplicate. Using the "Remove duplicates" that 5 Ids are removed. But the other 3 Ids are included in the data set.

	A	B	C	D	E	F	G	H
	Customer I	Date of Birth	Age	childre	charge	Hospital tie	City ti	State I
	Id2042	09-Jun-1999	23	1	$2,775.19	tier - 2	tier - 1	R1012
	Id2041	29-Aug-1999	23	1	$2,775.19	tier - 2	tier - 1	R1012
	Unknown 2	09-Jun-1999	23	1	$2,775.19	tier - 2	tier - 1	R1012

	A	B	C	D	E	F	G	H
	Customer I	Date of Birth	Age	childre	charge	Hospital tie	City ti	State I
2	Id1605	20-Dec-1985	37	2	$6,203.90	tier - 1	tier - 2	R1012
3	Id1604	22-Sep-1985	37	2	$6,203.90	tier - 2	tier - 2	R1012
41	Unknown 3	20-Dec-1985	37	2	$6,203.90	tier - 1	tier - 2	R1012

	A	B	C	D	E	F	G	H
	Customer I	Date of Birth	Age	childre	charge	Hospital tie	City ti	State I
0	Id209	13-Oct-2000	22	0	$35,585.58	tier - 1	tier - 2	R1011
1	Id208	19-Nov-2000	22	0	$35,585.58	tier - 2	tier - 3	R1011
1	Unknown 4	13-Oct-2000	22	0	$35,585.58	tier - 1	tier - 2	R1011

	A	B	C	D	E	F	G	H	I
1	Customer I	Date of Birth	Age	childre	charge	Hospital tie	City ti	State I	
2159	Id180	06-Oct-1992	30	0	$36,837.47	tier - 1	tier - 2	R1011	
2160	Id179	04-Jul-1992	30	0	$36,837.47	tier - 2	tier - 3	R1011	
2341	Unknown 5	06-Oct-1992	30	0	$36,837.47	tier - 1	tier - 2	R1011	
2343									
2344									

	A	B	C	D	E	F	G	H	I
	Customer I	Date of Birth	Age	childre	charge	Hospital tie	City ti	State I	
03	Id136	22-Nov-1991	31	2	$38,711.00	tier - 1	tier - 3	R1011	
04	Id135	02-Nov-1991	31	2	$38,711.00	tier - 2	tier - 2	R1011	
41	Unknown 6	22-Nov-1991	31	2	$38,711.00	tier - 1	tier - 3	R1011	
42									

Data imputation

The number of missing values marked with '?' in each column of the "Medical Examinations" table and "Hospitalization Details" tables are checked by clicking the "?" in the filter.

Customer Id: The missing customer Id is imputed as "Unknown".

	A	B	C	D	E	F	G	H
1	Customer	Date of Birth	Age	childre	charge	Hospital tie	City ti	State I
41	Unknown	06-Nov-2004	18	0	1137.01	tier - 3	tier - 1	R1013
42								

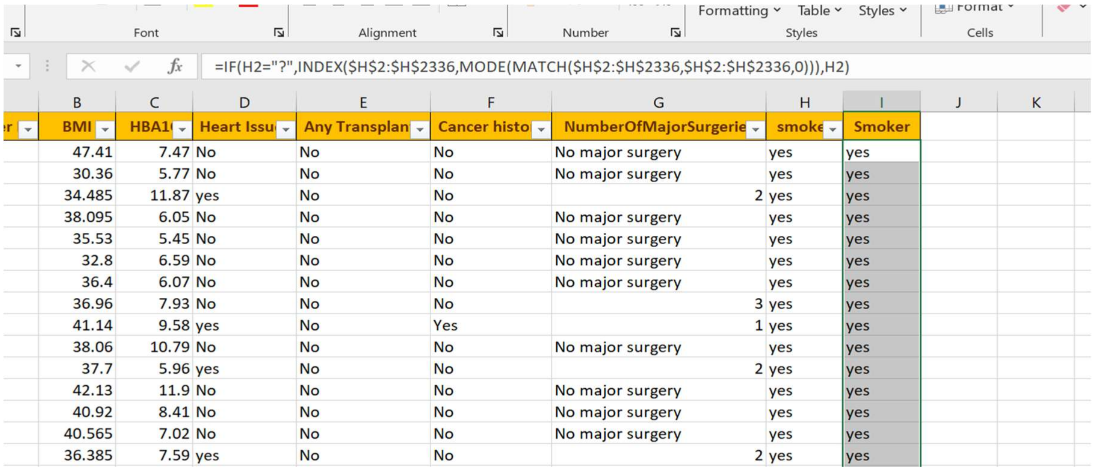
Smoker:

In medical examination sheet 2 smoker values are missing.

smoker

	A	B	C	D	E	F	G	H
	Customer	BMI	HBA1(	Heart Issu	Any Transplan	Cancer histo	NumberOfMajorSurgerie	smoke
1	Id560	23.98	4.9	No	No	No	No major surgery	?
6	Id635	25.175	4.96	No	yes	No	1	?

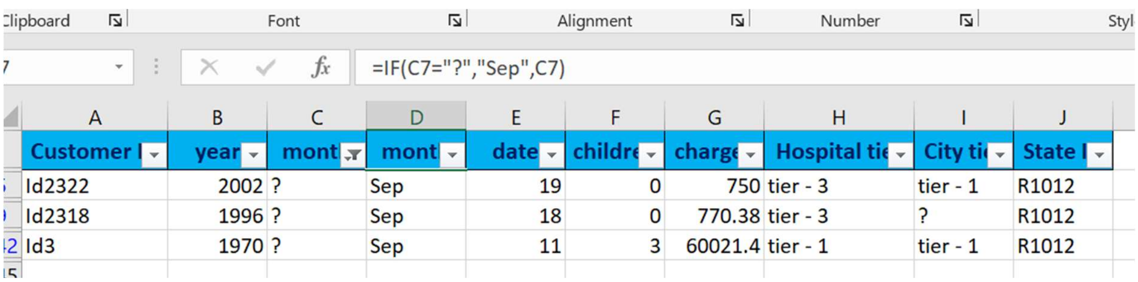
Based on the given instructions the missing values in the 'smoker' was imputed by the most frequently occurring values in the respective columns. Using the functions IF, INDEX, MODE and MATCH the missing values are imputed.



=IF(H2="?",INDEX($H$2:$H$2336,MODE(MATCH($H$2:$H$2336,$H$2:$H$2336,0))),H2)

BMI	HBA1(	Heart Issu	Any Transplan	Cancer histo	NumberOfMajorSurgerie	smoke	Smoker
47.41	7.47	No	No	No	No major surgery	yes	yes
30.36	5.77	No	No	No	No major surgery	yes	yes
34.485	11.87	yes	No	No	2	yes	yes
38.095	6.05	No	No	No	No major surgery	yes	yes
35.53	5.45	No	No	No	No major surgery	yes	yes
32.8	6.59	No	No	No	No major surgery	yes	yes
36.4	6.07	No	No	No	No major surgery	yes	yes
36.96	7.93	No	No	No	3	yes	yes
41.14	9.58	yes	No	Yes	1	yes	yes
38.06	10.79	No	No	No	No major surgery	yes	yes
37.7	5.96	yes	No	No	2	yes	yes
42.13	11.9	No	No	No	No major surgery	yes	yes
40.92	8.41	No	No	No	No major surgery	yes	yes
40.565	7.02	No	No	No	No major surgery	yes	yes
36.385	7.59	yes	No	No	2	yes	yes

Month: In Hospitalization Details 3 month values are missing. Based on the given instructions the missing values of 'month' was filled with "September" using the formula **=IF (C2= "?", "Sep", C2)**

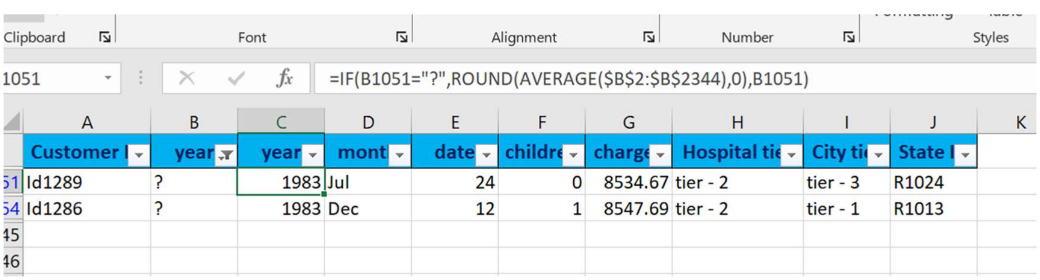


=IF(C7="?","Sep",C7)

Customer I	year	mont	mont	date	childre	charge	Hospital tie	City ti	State I
Id2322	2002	?	Sep	19	0	750	tier - 3	tier - 1	R1012
Id2318	1996	?	Sep	18	0	770.38	tier - 3	?	R1012
Id3	1970	?	Sep	11	3	60021.4	tier - 1	tier - 1	R1012

Year:

In Hospitalization Details, two year values are missing. Based the given instructions the missing year values are replaced with its average rounded value using the formula **=IF(B2= "?",ROUND(AVERAGE($B$2:$B$2344),0),B2)**



=IF(B1051="?",ROUND(AVERAGE($B$2:$B$2344),0),B1051)

Customer I	year	year	mont	date	childre	charge	Hospital tie	City ti	State I
Id1289	?	1983	Jul	24	0	8534.67	tier - 2	tier - 3	R1024
Id1286	?	1983	Dec	12	1	8547.69	tier - 2	tier - 1	R1013

Hospital tier and City tier:

The missing values in 'hospital tier', and 'city tier' are imputed by the most frequently occurring values in the respective columns by the given instructions. Using the functions IF, INDEX, MODE and MATCH the missing values are imputed.

=IF(G13="?",INDEX($G$2:$G$2344,MODE(MATCH($G$2:$G$2344,$G$2:$G$2344,0))),G13)

Year	Mont	date	childre	charge	Hospital tie	Hospital tie	City ti	State I
1999	Dec	26	0	700	?	tier - 2	tier - 3	R1013

=IF(G2="?",INDEX($G$2:$G$2344,MODE(MATCH($G$2:$G$2344,$G$2:$G$2344,0))),G2)

Year	Mont	date	childre	charge	Hospital tie	Hospital tie	City ti	State I
1992	Jul	9	0	563.84	tier - 2	tier - 2	tier - 3	R1013
1992	Nov	30	0	570.62	tier - 2	tier - 2	tier - 1	R1013
1993	Jun	30	0	600	tier - 2	tier - 2	tier - 1	R1013
1992	Sep	13	0	604.54	tier - 3	tier - 3	tier - 3	R1013
1998	Jul	27	0	637.26	tier - 3	tier - 3	tier - 3	R1013
2001	Nov	20	0	646.14	tier - 3	tier - 3	tier - 3	R1012
1993	Jun	1	0	650	tier - 3	tier - 3	tier - 3	R1013
1995	Jul	4	0	650	tier - 3	tier - 3	tier - 3	R1013
2002	Nov	29	0	668	tier - 3	tier - 3	tier - 2	R1012
1997	Nov	9	0	670	tier - 3	tier - 3	tier - 3	R1013
2001	Sep	12	0	687.54	tier - 3	tier - 3	tier - 2	R1013
1999	Dec	26	0	700	?	tier - 2	tier - 3	R1013
1999	Dec	14	0	722.99	tier - 3	tier - 3	tier - 1	R1013

=IF(H19="?",INDEX($H$2:$H$2344,MODE(MATCH($H$2:$H$2344,$H$2:$H$2344,0))),H19)

Year	Mont	date	childre	charge	Hospital tie	City ti	City ti	State I
1996	Sep	18	0	770.38	tier - 3	?	tier - 2	R1012

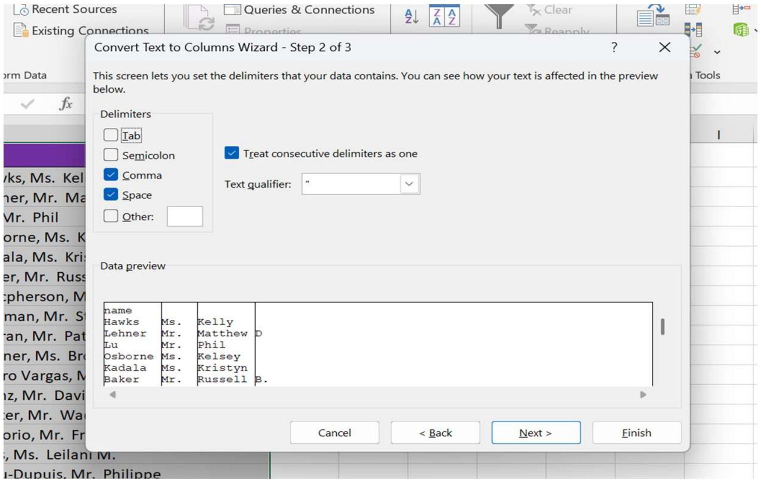
State Id: The missing State Ids are imputed as "Unknown" using the "IF" function based on the given instructions.

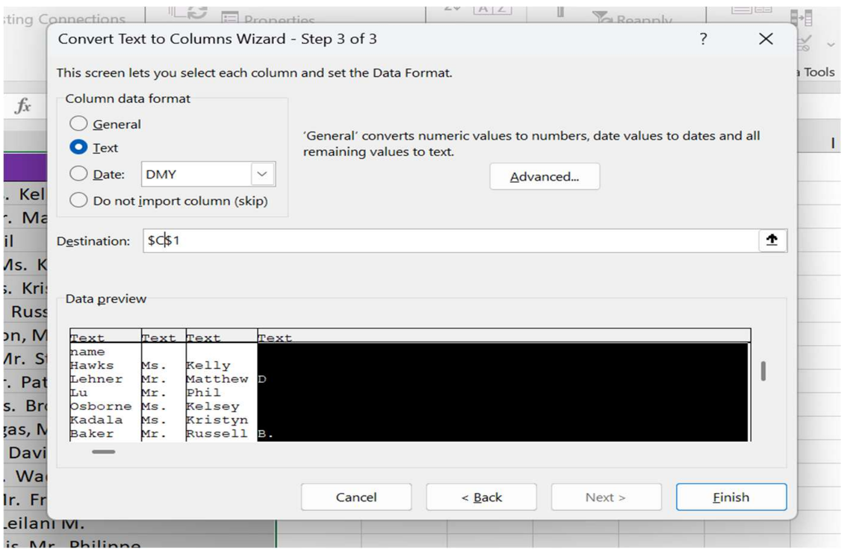
=IF(I546="?","Unknown",I546)

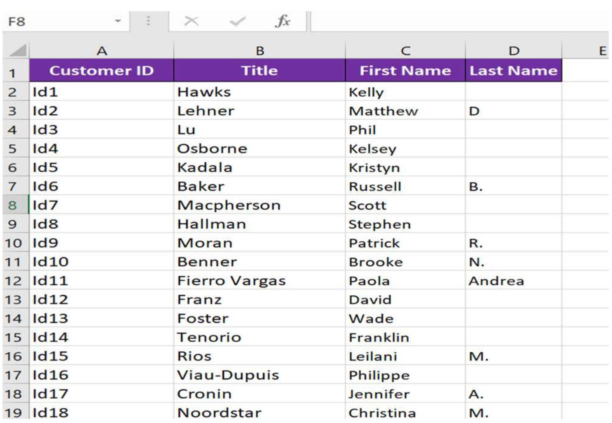
date	childre	charge	Hospital tie	City ti	State I	State I
1	3	4827.9	tier - 1	tier - 2	?	Unknown
5	1	37165.16	tier - 1	tier - 3	?	Unknown

Data Transformation:

The 'Names' column in the "Customer Names" table is split into 3 meaningful columns: 'Title', 'First Name', and 'Last Name' using the function **"Text to column".**

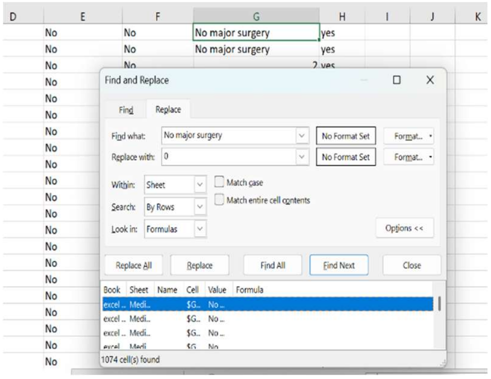


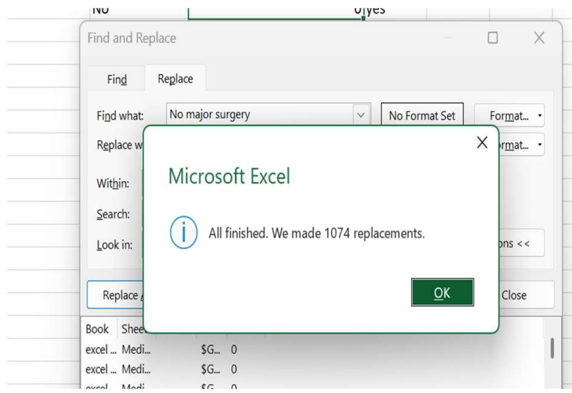




	A	B	C	D
1	Customer ID	Title	First Name	Last Name
2	Id1	Hawks	Kelly	
3	Id2	Lehner	Matthew	D
4	Id3	Lu	Phil	
5	Id4	Osborne	Kelsey	
6	Id5	Kadala	Kristyn	
7	Id6	Baker	Russell	B.
8	Id7	Macpherson	Scott	
9	Id8	Hallman	Stephen	
10	Id9	Moran	Patrick	R.
11	Id10	Benner	Brooke	N.
12	Id11	Fierro Vargas	Paola	Andrea
13	Id12	Franz	David	
14	Id13	Foster	Wade	
15	Id14	Tenorio	Franklin	
16	Id15	Rios	Leilani	M.
17	Id16	Viau-Dupuis	Philippe	
18	Id17	Cronin	Jennifer	A.
19	Id18	Noordstar	Christina	M.

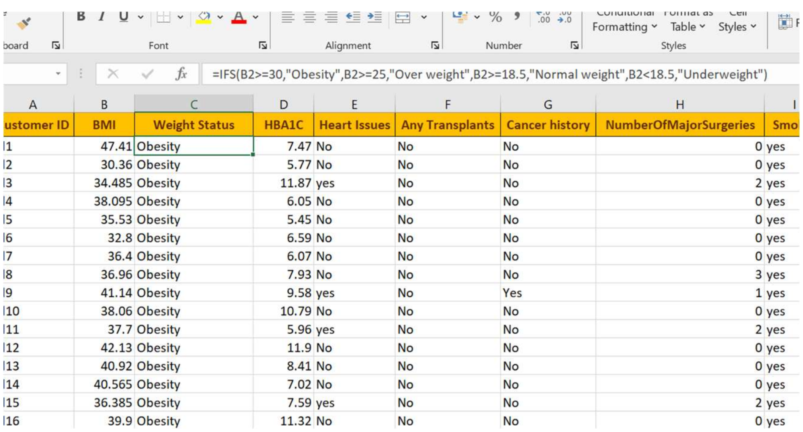
The text “no major surgery” in the column "Number of Major Surgeries" in "Medical Examinations" table is replaced by the numerical values 0 using “Find and Replace”





The new column named "Weight Status" was created using given below table and "IFS" function based on BMI values.

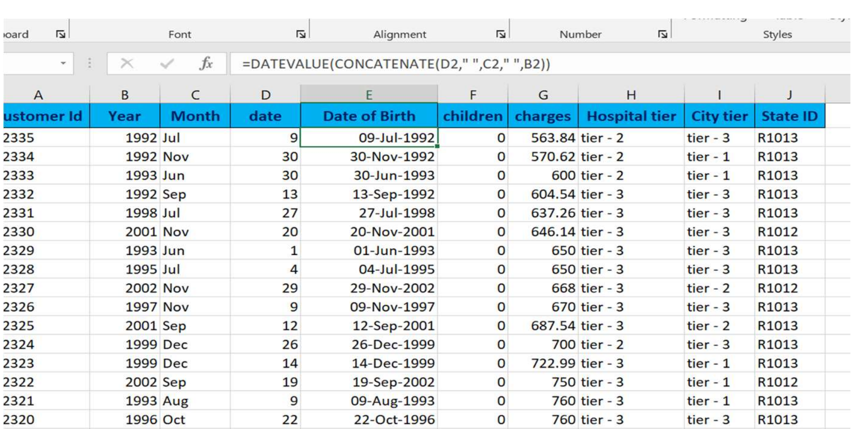
BMI	**Weight Status**
Below 18.5	Underweight
18.5 – 24.9	Normal Weight
25.0 – 29.9	Overweight
30.0 and Above	Obesity



=IFS(B2>=30,"Obesity",B2>=25,"Over weight",B2>=18.5,"Normal weight",B2<18.5,"Underweight")

ustomer ID	BMI	Weight Status	HBA1C	Heart Issues	Any Transplants	Cancer history	NumberOfMajorSurgeries	Smo
l1	47.41	Obesity	7.47	No	No	No	0	yes
l2	30.36	Obesity	5.77	No	No	No	0	yes
l3	34.485	Obesity	11.87	yes	No	No	2	yes
l4	38.095	Obesity	6.05	No	No	No	0	yes
l5	35.53	Obesity	5.45	No	No	No	0	yes
l6	32.8	Obesity	6.59	No	No	No	0	yes
l7	36.4	Obesity	6.07	No	No	No	0	yes
l8	36.96	Obesity	7.93	No	No	No	3	yes
l9	41.14	Obesity	9.58	yes	No	Yes	1	yes
l10	38.06	Obesity	10.79	No	No	No	0	yes
l11	37.7	Obesity	5.96	yes	No	No	2	yes
l12	42.13	Obesity	11.9	No	No	No	0	yes
l13	40.92	Obesity	8.41	No	No	No	0	yes
l14	40.565	Obesity	7.02	No	No	No	0	yes
l15	36.385	Obesity	7.59	yes	No	No	2	yes
l16	39.9	Obesity	11.32	No	No	No	0	yes

The new column 'Date of Birth' was created by merged the 'year', 'month' and 'date' columns in the "Hospitalization Details" table using the formula **=DATEVALUE(CONCATENATE(D2, " ",C2, " ",B2)).** The custom format 'DD-MMM-YYYY' is used in this column.



=DATEVALUE(CONCATENATE(D2," ",C2," ",B2))

ustomer Id	Year	Month	date	Date of Birth	children	charges	Hospital tier	City tier	State ID
2335	1992	Jul	9	09-Jul-1992	0	563.84	tier - 2	tier - 3	R1013
2334	1992	Nov	30	30-Nov-1992	0	570.62	tier - 2	tier - 1	R1013
2333	1993	Jun	30	30-Jun-1993	0	600	tier - 2	tier - 1	R1013
2332	1992	Sep	13	13-Sep-1992	0	604.54	tier - 3	tier - 3	R1013
2331	1998	Jul	27	27-Jul-1998	0	637.26	tier - 3	tier - 3	R1013
2330	2001	Nov	20	20-Nov-2001	0	646.14	tier - 3	tier - 3	R1012
2329	1993	Jun	1	01-Jun-1993	0	650	tier - 3	tier - 3	R1013
2328	1995	Jul	4	04-Jul-1995	0	650	tier - 3	tier - 3	R1013
2327	2002	Nov	29	29-Nov-2002	0	668	tier - 3	tier - 2	R1012
2326	1997	Nov	9	09-Nov-1997	0	670	tier - 3	tier - 3	R1013
2325	2001	Sep	12	12-Sep-2001	0	687.54	tier - 3	tier - 2	R1013
2324	1999	Dec	26	26-Dec-1999	0	700	tier - 2	tier - 3	R1013
2323	1999	Dec	14	14-Dec-1999	0	722.99	tier - 3	tier - 1	R1013
2322	2002	Sep	19	19-Sep-2002	0	750	tier - 3	tier - 1	R1012
2321	1993	Aug	9	09-Aug-1993	0	760	tier - 3	tier - 1	R1013
2320	1996	Oct	22	22-Oct-1996	0	760	tier - 3	tier - 3	R1013
2319	1993	Jun	28	28-Jun-1993	0	770	tier - 3	tier - 3	R1013

The ‘Age’ of each customer based on their ‘Date of Birth’ and the date of collection of the dataset, which is 8th June 2023 was calculated by using the formula “DATEDIF” and “Y”

=DATEDIF(B2,$K$3,"Y")

omer	Date of Birth	Age	childre	charge	Hospital tie	City ti	State I	I	J	K
5	09-Jul-1992	30	0	563.84	tier - 2	tier - 3	R1013			
4	30-Nov-1992	30	0	570.62	tier - 2	tier - 1	R1013			08-Jun-23
3	30-Jun-1993	29	0	600	tier - 2	tier - 1	R1013			
2	13-Sep-1992	30	0	604.54	tier - 3	tier - 3	R1013			
1	27-Jul-1998	24	0	637.26	tier - 3	tier - 3	R1013			
0	20-Nov-2001	21	0	646.14	tier - 3	tier - 3	R1012			
9	01-Jun-1993	30	0	650	tier - 3	tier - 3	R1013			
8	04-Jul-1995	27	0	650	tier - 3	tier - 3	R1013			
7	29-Nov-2002	20	0	668	tier - 3	tier - 2	R1012			
6	09-Nov-1997	25	0	670	tier - 3	tier - 3	R1013			
5	12-Sep-2001	21	0	687.54	tier - 3	tier - 2	R1013			
4	26-Dec-1999	23	0	700	tier - 2	tier - 3	R1013			
3	14-Dec-1999	23	0	722.99	tier - 3	tier - 1	R1013			
2	19-Sep-2002	20	0	750	tier - 3	tier - 1	R1012			
1	09-Aug-1993	29	0	760	tier - 3	tier - 1	R1013			
0	22-Oct-1996	26	0	760	tier - 3	tier - 3	R1013			
9	28-Jun-1993	29	0	770	tier - 3	tier - 3	R1013			
8	18-Sep-1996	26	0	770.38	tier - 3	tier - 2	R1012			
7	07-Dec-1995	27	0	773.54	tier - 3	tier - 2	R1013			
6	07-Oct-2004	18	0	830.52	tier - 3	tier - 2	R1011			

The ‘Charges’ column was formated into currency ($).

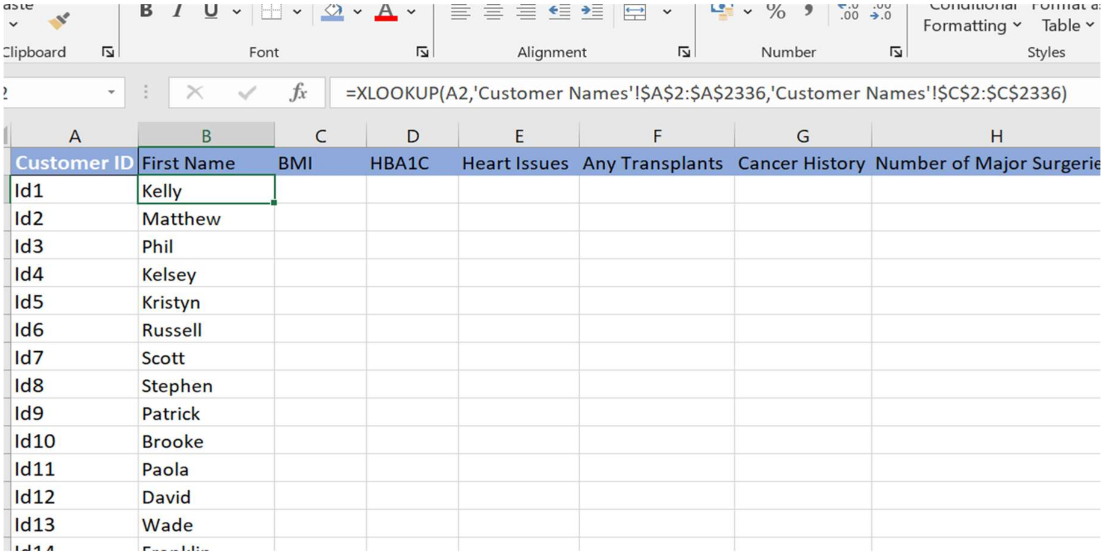
ge	children	charges	Hospital tier	City tier	State ID
30	0	$563.84	tier - 2	tier - 3	R1013
30	0	$570.62	tier - 2	tier - 1	R1013
29	0	$600.00	tier - 2	tier - 1	R1013
30	0	$604.54	tier - 3	tier - 3	R1013
24	0	$637.26	tier - 3	tier - 3	R1013
21	0	$646.14	tier - 3	tier - 3	R1012
30	0	$650.00	tier - 3	tier - 3	R1013
27	0	$650.00	tier - 3	tier - 3	R1013
20	0	$668.00	tier - 3	tier - 2	R1012
25	0	$670.00	tier - 3	tier - 3	R1013
21	0	$687.54	tier - 3	tier - 2	R1013
23	0	$700.00	tier - 2	tier - 3	R1013
23	0	$722.99	tier - 3	tier - 1	R1013
20	0	$750.00	tier - 3	tier - 1	R1012

AI-Driven Data Exploration, Analysis & Visualization:

Data Preparation:

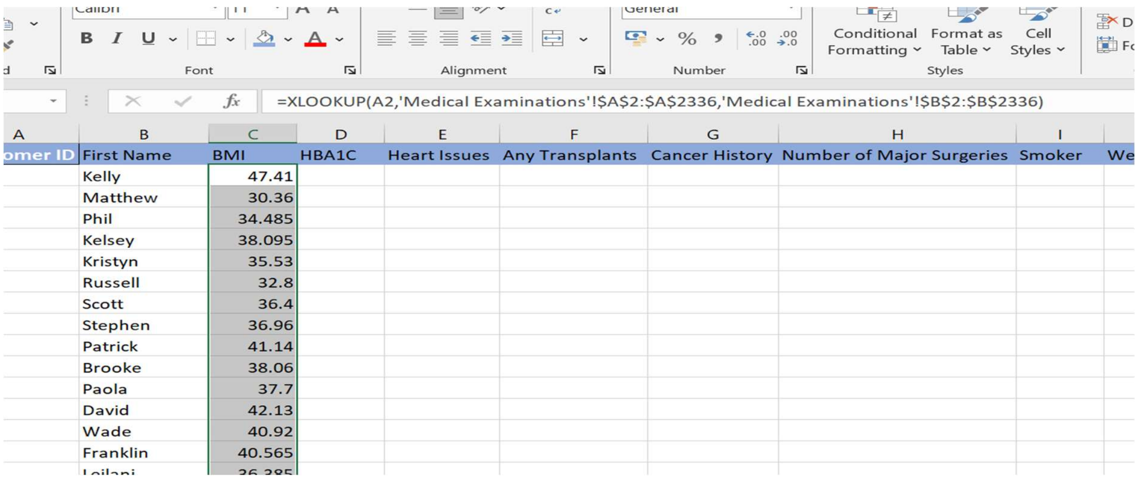
A new sheet named **“Healthcare”** was created using the above three table using the common column “Customer ID”. With the help of XLOOKUP formula the healthcare sheet was created. It contains the following columns, Customer ID, First Name, BMI, HBA1C, Heart Issues, Any Transplants, Cancer History, Number of Major Surgeries, Smoker, Weight Status, Diabetes Status, Date of Birth, Charges, Hospital Tier, City Tier, State ID, and Age. The diabetic status was filled with “**IFS**” function based on the “HBA1C” range given below.

HBA1C	**Diabetic status**
Normal	Below 5.7
prediabetes	5.7-6.4
Diabetes	Above 6.5



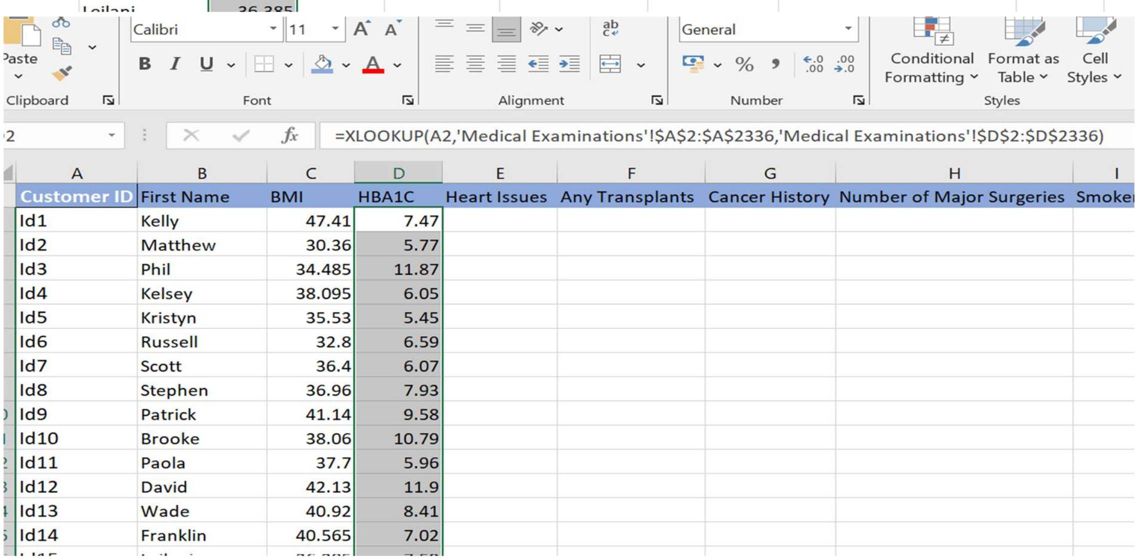
=XLOOKUP(A2,'Customer Names'!$A$2:$A$2336,'Customer Names'!$C$2:$C$2336)

Customer ID	First Name	BMI	HBA1C	Heart Issues	Any Transplants	Cancer History	Number of Major Surgerie
Id1	Kelly						
Id2	Matthew						
Id3	Phil						
Id4	Kelsey						
Id5	Kristyn						
Id6	Russell						
Id7	Scott						
Id8	Stephen						
Id9	Patrick						
Id10	Brooke						
Id11	Paola						
Id12	David						
Id13	Wade						



=XLOOKUP(A2,'Medical Examinations'!$A$2:$A$2336,'Medical Examinations'!$B$2:$B$2336)

omer ID	First Name	BMI	HBA1C	Heart Issues	Any Transplants	Cancer History	Number of Major Surgeries	Smoker	We
	Kelly	47.41							
	Matthew	30.36							
	Phil	34.485							
	Kelsey	38.095							
	Kristyn	35.53							
	Russell	32.8							
	Scott	36.4							
	Stephen	36.96							
	Patrick	41.14							
	Brooke	38.06							
	Paola	37.7							
	David	42.13							
	Wade	40.92							
	Franklin	40.565							



=XLOOKUP(A2,'Medical Examinations'!$A$2:$A$2336,'Medical Examinations'!$D$2:$D$2336)

Customer ID	First Name	BMI	HBA1C	Heart Issues	Any Transplants	Cancer History	Number of Major Surgeries	Smoker
Id1	Kelly	47.41	7.47					
Id2	Matthew	30.36	5.77					
Id3	Phil	34.485	11.87					
Id4	Kelsey	38.095	6.05					
Id5	Kristyn	35.53	5.45					
Id6	Russell	32.8	6.59					
Id7	Scott	36.4	6.07					
Id8	Stephen	36.96	7.93					
Id9	Patrick	41.14	9.58					
Id10	Brooke	38.06	10.79					
Id11	Paola	37.7	5.96					
Id12	David	42.13	11.9					
Id13	Wade	40.92	8.41					
Id14	Franklin	40.565	7.02					

=XLOOKUP(A2,'Medical Examinations'!$A$2:$A$2336,'Medical Examinations'!$E$2:$E$2336)

Customer ID	First Name	BMI	HBA1C	Heart Issues	Any Transplants	Cancer History	Number of Major Surgeries	Smoker
Id1	Kelly	47.41	7.47	No				
Id2	Matthew	30.36	5.77	No				
Id3	Phil	34.485	11.87	yes				
Id4	Kelsey	38.095	6.05	No				
Id5	Kristyn	35.53	5.45	No				
Id6	Russell	32.8	6.59	No				
Id7	Scott	36.4	6.07	No				
Id8	Stephen	36.96	7.93	No				
Id9	Patrick	41.14	9.58	yes				
Id10	Brooke	38.06	10.79	No				
Id11	Paola	37.7	5.96	yes				
Id12	David	42.13	11.9	No				
Id13	Wade	40.92	8.41	No				
Id14	Franklin	40.565	7.02	No				

=XLOOKUP(A2,'Medical Examinations'!$A$2:$A$2336,'Medical Examinations'!$F$2:$F$2336)

Customer ID	First Name	BMI	HBA1C	Heart Issues	Any Transplants	Cancer History	Number of Major Surgeries	Smoker
Id1	Kelly	47.41	7.47	No	No			
Id2	Matthew	30.36	5.77	No	No			
Id3	Phil	34.485	11.87	yes	No			
Id4	Kelsey	38.095	6.05	No	No			
Id5	Kristyn	35.53	5.45	No	No			
Id6	Russell	32.8	6.59	No	No			
Id7	Scott	36.4	6.07	No	No			
Id8	Stephen	36.96	7.93	No	No			
Id9	Patrick	41.14	9.58	yes	No			
Id10	Brooke	38.06	10.79	No	No			
Id11	Paola	37.7	5.96	yes	No			
Id12	David	42.13	11.9	No	No			

=XLOOKUP(A2,'Medical Examinations'!$A$2:$A$2336,'Medical Examinations'!$G$2:$G$2336)

Customer ID	First Name	BMI	HBA1C	Heart Issues	Any Transplants	Cancer History	Number of Major Surgeries	Smoker
Id1	Kelly	47.41	7.47	No	No	No		
Id2	Matthew	30.36	5.77	No	No	No		
Id3	Phil	34.485	11.87	yes	No	No		
Id4	Kelsey	38.095	6.05	No	No	No		
Id5	Kristyn	35.53	5.45	No	No	No		
Id6	Russell	32.8	6.59	No	No	No		
Id7	Scott	36.4	6.07	No	No	No		
Id8	Stephen	36.96	7.93	No	No	No		
Id9	Patrick	41.14	9.58	yes	No	Yes		
Id10	Brooke	38.06	10.79	No	No	No		
Id11	Paola	37.7	5.96	yes	No	No		
Id12	David	42.13	11.9	No	No	No		
Id13	Wade	40.92	8.41	No	No	No		

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=XLOOKUP(A2,'Medical Examinations'!$A$2:$A$2336,'Medical Examinations'!$I$2:$I$2336)

Customer ID	First Name	BMI	HBA1C	Heart Issues	Any Transplants	Cancer History	Number of Major Surgeries	Smoker	Weight Statu
Id1	Kelly	47.41	7.47	No	No	No	0	yes	
Id2	Matthew	30.36	5.77	No	No	No	0	yes	
Id3	Phil	34.485	11.87	yes	No	No	2	yes	
Id4	Kelsey	38.095	6.05	No	No	No	0	yes	
Id5	Kristyn	35.53	5.45	No	No	No	0	yes	
Id6	Russell	32.8	6.59	No	No	No	0	yes	
Id7	Scott	36.4	6.07	No	No	No	0	yes	
Id8	Stephen	36.96	7.93	No	No	No	3	yes	
Id9	Patrick	41.14	9.58	yes	No	Yes	1	yes	
Id10	Brooke	38.06	10.79	No	No	No	0	yes	
Id11	Paola	37.7	5.96	yes	No	No	2	yes	
Id12	David	42.13	11.9	No	No	No	0	yes	
Id13	Wade	40.92	8.41	No	No	No	0	yes	
Id14	Franklin	40.565	7.02	No	No	No	0	yes	
Id15	Leilani	36.385	7.59	yes	No	No	2	yes	

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=XLOOKUP(A2,'Medical Examinations'!$A$2:$A$2336,'Medical Examinations'!$C$2:$C$2336)

Customer ID	First Name	BMI	HBA1C	Heart Issues	Any Transplants	Cancer History	Number of Major Surgeries	Smoker	Weight Status	Diabetes Stat
Id1	Kelly	47.41	7.47	No	No	No	0	yes	Obesity	
Id2	Matthew	30.36	5.77	No	No	No	0	yes	Obesity	
Id3	Phil	34.485	11.87	yes	No	No	2	yes	Obesity	
Id4	Kelsey	38.095	6.05	No	No	No	0	yes	Obesity	
Id5	Kristyn	35.53	5.45	No	No	No	0	yes	Obesity	
Id6	Russell	32.8	6.59	No	No	No	0	yes	Obesity	
Id7	Scott	36.4	6.07	No	No	No	0	yes	Obesity	
Id8	Stephen	36.96	7.93	No	No	No	3	yes	Obesity	
Id9	Patrick	41.14	9.58	yes	No	Yes	1	yes	Obesity	
Id10	Brooke	38.06	10.79	No	No	No	0	yes	Obesity	
Id11	Paola	37.7	5.96	yes	No	No	2	yes	Obesity	
Id12	David	42.13	11.9	No	No	No	0	yes	Obesity	
Id13	Wade	40.92	8.41	No	No	No	0	yes	Obesity	
Id14	Franklin	40.565	7.02	No	No	No	0	yes	Obesity	
Id15	Leilani	36.385	7.59	yes	No	No	2	yes	Obesity	
Id16	Philippe	39.9	11.32	No	No	No	0	yes	Obesity	
Id17	Jennifer	33.8	7.67	No	No	No	3	yes	Obesity	
Id18	Christina	36.765	7.29	yes	No	Yes	1	yes	Obesity	

Clipboard | Font | Alignment | Number | Conditional Formatting | Format as Table | Cell Styles | Styles | Delete | Format | Cells | Sort & Filter | Find & Select | Editing

=IFS(D2>=6.5,"Diabetes",D2>=5.7,"Prediabetes",D2<5.7,"Normal")

Customer ID	First Name	BMI	HBA1C	Heart Issues	Any Transplants	Cancer History	Number of Major Surgeries	Smoker	Weight Status	Diabetes Status	Date of B
Id1	Kelly	47.41	7.47	No	No	No	0	yes	Obesity	Diabetes	
Id2	Matthew	30.36	5.77	No	No	No	0	yes	Obesity	Prediabetes	
Id3	Phil	34.485	11.87	yes	No	No	2	yes	Obesity	Diabetes	
Id4	Kelsey	38.095	6.05	No	No	No	0	yes	Obesity	Prediabetes	
Id5	Kristyn	35.53	5.45	No	No	No	0	yes	Obesity	Normal	
Id6	Russell	32.8	6.59	No	No	No	0	yes	Obesity	Diabetes	
Id7	Scott	36.4	6.07	No	No	No	0	yes	Obesity	Prediabetes	
Id8	Stephen	36.96	7.93	No	No	No	3	yes	Obesity	Diabetes	
Id9	Patrick	41.14	9.58	yes	No	Yes	1	yes	Obesity	Diabetes	
Id10	Brooke	38.06	10.79	No	No	No	0	yes	Obesity	Diabetes	
Id11	Paola	37.7	5.96	yes	No	No	2	yes	Obesity	Prediabetes	
Id12	David	42.13	11.9	No	No	No	0	yes	Obesity	Diabetes	
Id13	Wade	40.92	8.41	No	No	No	0	yes	Obesity	Diabetes	
Id14	Franklin	40.565	7.02	No	No	No	0	yes	Obesity	Diabetes	
Id15	Leilani	36.385	7.59	yes	No	No	2	yes	Obesity	Diabetes	
Id16	Philippe	39.9	11.32	No	No	No	0	yes	Obesity	Diabetes	
Id17	Jennifer	33.8	7.67	No	No	No	3	yes	Obesity	Diabetes	

=XLOOKUP(A2,'Hospitalisation Details'!$A$2:$A$2344,'Hospitalisation Details'!$B$2:$B$2344)

me	BMI	HBA1C	Heart Issues	Any Transplants	Cancer History	Number of Major Surgeries	Smoker	Weight Status	Diabetes Status	Date of Birth	Charges
	47.41	7.47	No	No	No	0	yes	Obesity	Diabetes	12 Oct 1968	
w	30.36	5.77	No	No	No	0	yes	Obesity	Prediabetes	08 Jun 1977	
	34.485	11.87	yes	No	No	2	yes	Obesity	Diabetes	11 Sep 1970	
	38.095	6.05	No	No	No	0	yes	Obesity	Prediabetes	06 Jun 1991	
	35.53	5.45	No	No	No	0	yes	Obesity	Normal	19 Jun 1989	
	32.8	6.59	No	No	No	0	yes	Obesity	Diabetes	04 Aug 1962	
	36.4	6.07	No	No	No	0	yes	Obesity	Prediabetes	27 Oct 1994	
n	36.96	7.93	No	No	No	3	yes	Obesity	Diabetes	27 Jun 1958	
	41.14	9.58	yes	No	Yes	1	yes	Obesity	Diabetes	04 Sep 1963	
	38.06	10.79	No	No	No	0	yes	Obesity	Diabetes	29 Dec 1978	
	37.7	5.96	yes	No	No	2	yes	Obesity	Prediabetes	22 Jul 1959	
	42.13	11.9	No	No	No	0	yes	Obesity	Diabetes	27 Oct 1965	
	40.92	8.41	No	No	No	0	yes	Obesity	Diabetes	11 Oct 1962	
ı	40.565	7.02	No	No	No	0	yes	Obesity	Diabetes	01 Dec 1968	
	36.385	7.59	yes	No	No	2	yes	Obesity	Diabetes	21 Dec 1961	

=XLOOKUP(A2,'Hospitalisation Details'!$A$2:$A$2344,'Hospitalisation Details'!$E$2:$E$2344)

	HBA1C	Heart Issues	Any Transplants	Cancer History	Number of Major Surgeries	Smoker	Weight Status	Diabetes Status	Date of Birth	Charges	Hospital
.41	7.47	No	No	No	0	yes	Obesity	Diabetes	12-Oct-1968	$63,770.43	
.36	5.77	No	No	No	0	yes	Obesity	Prediabetes	08-Jun-1977	$62,592.87	
185	11.87	yes	No	No	2	yes	Obesity	Diabetes	11-Sep-1970	$60,021.40	
)95	6.05	No	No	No	0	yes	Obesity	Prediabetes	06-Jun-1991	$58,571.07	
.53	5.45	No	No	No	0	yes	Obesity	Normal	19-Jun-1989	$55,135.40	
2.8	6.59	No	No	No	0	yes	Obesity	Diabetes	04-Aug-1962	$52,590.83	
6.4	6.07	No	No	No	0	yes	Obesity	Prediabetes	27-Oct-1994	$51,194.56	
.96	7.93	No	No	No	3	yes	Obesity	Diabetes	27-Jun-1958	$49,577.66	
.14	9.58	yes	No	Yes	1	yes	Obesity	Diabetes	04-Sep-1963	$48,970.25	
.06	10.79	No	No	No	0	yes	Obesity	Diabetes	29-Dec-1978	$48,885.14	
7.7	5.96	yes	No	No	2	yes	Obesity	Prediabetes	22-Jul-1959	$48,824.45	
.13	11.9	No	No	No	0	yes	Obesity	Diabetes	27-Oct-1965	$48,675.52	
.92	8.41	No	No	No	0	yes	Obesity	Diabetes	11-Oct-1962	$48,673.56	
565	7.02	No	No	No	0	yes	Obesity	Diabetes	01-Dec-1968	$48,549.18	
385	7.59	yes	No	No	2	yes	Obesity	Diabetes	21-Dec-1961	$48,517.56	
9.9	11.32	No	No	No	0	yes	Obesity	Diabetes	27-Aug-1962	$48,173.36	

=XLOOKUP(A2,'Hospitalisation Details'!$A$2:$A$2344,'Hospitalisation Details'!$F$2:$F$2344)

HBA1C	Heart Issues	Any Transplants	Cancer History	Number of Major Surgeries	Smoker	Weight Status	Diabetes Status	Date of Birth	Charges	Hospital Tier	City Tier
7.47	No	No	No	0	yes	Obesity	Diabetes	12-Oct-1968	$63,770.43	tier - 1	
5.77	No	No	No	0	yes	Obesity	Prediabetes	08-Jun-1977	$62,592.87	tier - 2	
11.87	yes	No	No	2	yes	Obesity	Diabetes	11-Sep-1970	$60,021.40	tier - 1	
6.05	No	No	No	0	yes	Obesity	Prediabetes	06-Jun-1991	$58,571.07	tier - 1	
5.45	No	No	No	0	yes	Obesity	Normal	19-Jun-1989	$55,135.40	tier - 1	
6.59	No	No	No	0	yes	Obesity	Diabetes	04-Aug-1962	$52,590.83	tier - 1	
6.07	No	No	No	0	yes	Obesity	Prediabetes	27-Oct-1994	$51,194.56	tier - 1	
7.93	No	No	No	3	yes	Obesity	Diabetes	27-Jun-1958	$49,577.66	tier - 2	
9.58	yes	No	Yes	1	yes	Obesity	Diabetes	04-Sep-1963	$48,970.25	tier - 1	
10.79	No	No	No	0	yes	Obesity	Diabetes	29-Dec-1978	$48,885.14	tier - 1	
5.96	yes	No	No	2	yes	Obesity	Prediabetes	22-Jul-1959	$48,824.45	tier - 2	
11.9	No	No	No	0	yes	Obesity	Diabetes	27-Oct-1965	$48,675.52	tier - 1	
8.41	No	No	No	0	yes	Obesity	Diabetes	11-Oct-1962	$48,673.56	tier - 1	
7.02	No	No	No	0	yes	Obesity	Diabetes	01-Dec-1968	$48,549.18	tier - 1	
7.59	yes	No	No	2	yes	Obesity	Diabetes	21-Dec-1961	$48,517.56	tier - 1	
11.32	No	No	No	0	yes	Obesity	Diabetes	27-Aug-1962	$48,173.36	tier - 1	
7.67	No	No	No	3	yes	Obesity	Diabetes	16-Nov-1958	$47,928.03	tier - 2	
7.29	yes	No	Yes	1	yes	Obesity	Diabetes	05-Aug-1963	$47,896.79	tier - 1	

=XLOOKUP(A2,'Hospitalisation Details'!$A$2:$A$2344,'Hospitalisation Details'!$G$2:$G$2344)

Any Transplants	Cancer History	Number of Major Surgeries	Smoker	Weight Status	Diabetes Status	Date of Birth	Charges	Hospital Tier	City Tier	State ID
No	No	0	yes	Obesity	Diabetes	12-Oct-1968	$63,770.43	tier - 1	tier - 3	
No	No	0	yes	Obesity	Prediabetes	08-Jun-1977	$62,592.87	tier - 2	tier - 3	
No	No	2	yes	Obesity	Diabetes	11-Sep-1970	$60,021.40	tier - 1	tier - 1	
No	No	0	yes	Obesity	Prediabetes	06-Jun-1991	$58,571.07	tier - 1	tier - 3	
No	No	0	yes	Obesity	Normal	19-Jun-1989	$55,135.40	tier - 1	tier - 2	
No	No	0	yes	Obesity	Diabetes	04-Aug-1962	$52,590.83	tier - 1	tier - 3	
No	No	0	yes	Obesity	Prediabetes	27-Oct-1994	$51,194.56	tier - 1	tier - 3	
No	No	3	yes	Obesity	Diabetes	27-Jun-1958	$49,577.66	tier - 2	tier - 2	
No	Yes	1	yes	Obesity	Diabetes	04-Sep-1963	$48,970.25	tier - 1	tier - 2	
No	No	0	yes	Obesity	Diabetes	29-Dec-1978	$48,885.14	tier - 1	tier - 2	
No	No	2	yes	Obesity	Prediabetes	22-Jul-1959	$48,824.45	tier - 2	tier - 1	
No	No	0	yes	Obesity	Diabetes	27-Oct-1965	$48,675.52	tier - 1	tier - 2	
No	No	0	yes	Obesity	Diabetes	11-Oct-1962	$48,673.56	tier - 1	tier - 2	
No	No	0	yes	Obesity	Diabetes	01-Dec-1968	$48,549.18	tier - 1	tier - 3	
No	No	2	yes	Obesity	Diabetes	21-Dec-1961	$48,517.56	tier - 1	tier - 3	
No	No	0	yes	Obesity	Diabetes	27-Aug-1962	$48,173.36	tier - 1	tier - 3	
No	No	3	yes	Obesity	Diabetes	16-Nov-1958	$47,928.03	tier - 2	tier - 3	

=XLOOKUP(A2,'Hospitalisation Details'!$A$2:$A$2344,'Hospitalisation Details'!$H$2:$H$2344)

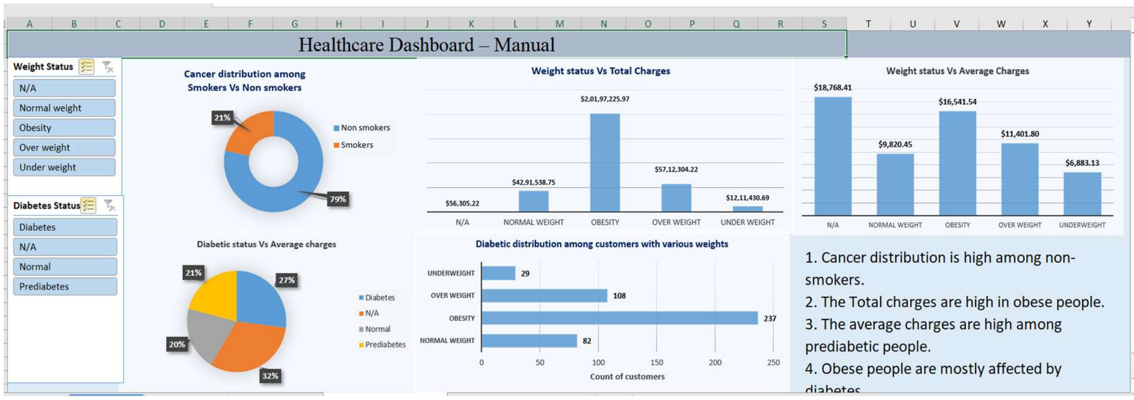
Any Transplants	Cancer History	Number of Major Surgeries	Smoker	Weight Status	Diabetes Status	Date of Birth	Charges	Hospital Tier	City Tier	State ID	Age
No	No	0	yes	Obesity	Diabetes	12-Oct-1968	$63,770.43	tier - 1	tier - 3	R1013	
No	No	0	yes	Obesity	Prediabetes	08-Jun-1977	$62,592.87	tier - 2	tier - 3	R1013	
No	No	2	yes	Obesity	Diabetes	11-Sep-1970	$60,021.40	tier - 1	tier - 1	R1012	
No	No	0	yes	Obesity	Prediabetes	06-Jun-1991	$58,571.07	tier - 1	tier - 3	R1024	
No	No	0	yes	Obesity	Normal	19-Jun-1989	$55,135.40	tier - 1	tier - 2	R1012	
No	No	0	yes	Obesity	Diabetes	04-Aug-1962	$52,590.83	tier - 1	tier - 3	R1011	
No	No	0	yes	Obesity	Prediabetes	27-Oct-1994	$51,194.56	tier - 1	tier - 3	R1011	
No	No	3	yes	Obesity	Diabetes	27-Jun-1958	$49,577.66	tier - 2	tier - 2	R1013	
No	Yes	1	yes	Obesity	Diabetes	04-Sep-1963	$48,970.25	tier - 1	tier - 2	R1013	
No	No	0	yes	Obesity	Diabetes	29-Dec-1978	$48,885.14	tier - 1	tier - 2	R1013	
No	No	2	yes	Obesity	Prediabetes	22-Jul-1959	$48,824.45	tier - 2	tier - 1	R1011	
No	No	0	yes	Obesity	Diabetes	27-Oct-1965	$48,675.52	tier - 1	tier - 2	R1013	
No	No	0	yes	Obesity	Diabetes	11-Oct-1962	$48,673.56	tier - 1	tier - 2	R1013	
No	No	0	yes	Obesity	Diabetes	01-Dec-1968	$48,549.18	tier - 1	tier - 3	R1016	
No	No	2	yes	Obesity	Diabetes	21-Dec-1961	$48,517.56	tier - 1	tier - 3	R1024	
No	No	0	yes	Obesity	Diabetes	27-Aug-1962	$48,173.36	tier - 1	tier - 3	R1011	

=XLOOKUP(A2,'Hospitalisation Details'!$A$2:$A$2344,'Hospitalisation Details'!$C$2:$C$2344)

Any Transplants	Cancer History	Number of Major Surgeries	Smoker	Weight Status	Diabetes Status	Date of Birth	Charges	Hospital Tier	City Tier	State ID	Age
No	No	0	yes	Obesity	Diabetes	12-Oct-1968	$63,770.43	tier - 1	tier - 3	R1013	54
No	No	0	yes	Obesity	Prediabetes	08-Jun-1977	$62,592.87	tier - 2	tier - 3	R1013	46
No	No	2	yes	Obesity	Diabetes	11-Sep-1970	$60,021.40	tier - 1	tier - 1	R1012	52
No	No	0	yes	Obesity	Prediabetes	06-Jun-1991	$58,571.07	tier - 1	tier - 3	R1024	32
No	No	0	yes	Obesity	Normal	19-Jun-1989	$55,135.40	tier - 1	tier - 2	R1012	33
No	No	0	yes	Obesity	Diabetes	04-Aug-1962	$52,590.83	tier - 1	tier - 3	R1011	60
No	No	0	yes	Obesity	Prediabetes	27-Oct-1994	$51,194.56	tier - 1	tier - 3	R1011	28
No	No	3	yes	Obesity	Diabetes	27-Jun-1958	$49,577.66	tier - 2	tier - 2	R1013	64
No	Yes	1	yes	Obesity	Diabetes	04-Sep-1963	$48,970.25	tier - 1	tier - 2	R1013	59
No	No	0	yes	Obesity	Diabetes	29-Dec-1978	$48,885.14	tier - 1	tier - 2	R1013	44
No	No	2	yes	Obesity	Prediabetes	22-Jul-1959	$48,824.45	tier - 2	tier - 1	R1011	63
No	No	0	yes	Obesity	Diabetes	27-Oct-1965	$48,675.52	tier - 1	tier - 2	R1013	57
No	No	0	yes	Obesity	Diabetes	11-Oct-1962	$48,673.56	tier - 1	tier - 2	R1013	60
No	No	0	yes	Obesity	Diabetes	01-Dec-1968	$48,549.18	tier - 1	tier - 3	R1016	54
No	No	2	yes	Obesity	Diabetes	21-Dec-1961	$48,517.56	tier - 1	tier - 3	R1024	61
No	No	0	yes	Obesity	Diabetes	27-Aug-1962	$48,173.36	tier - 1	tier - 3	R1011	60

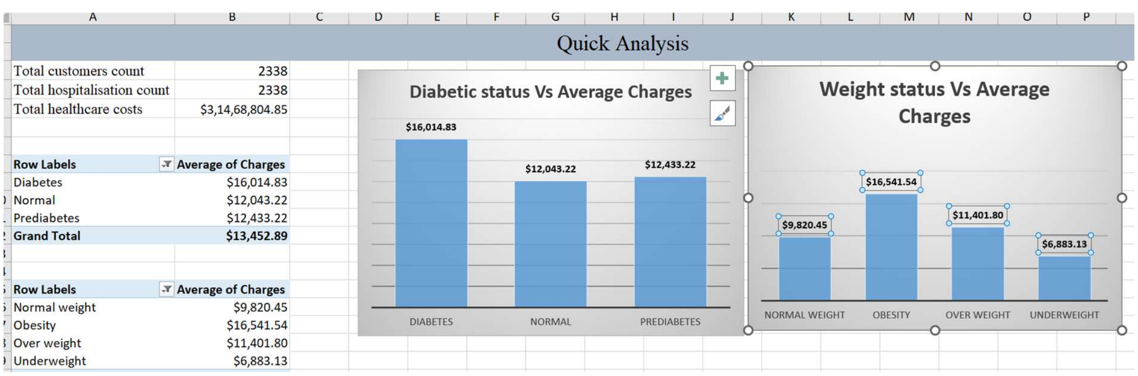
Manual Dashboard

Pivot tables are created manually using the combined healthcare dataset. A donut chart was created based on cancer distibution among smokers Vs non-smokers. The clustered column chart was created based on charges by weight status. Diabetic distribution among the customers based on various weight and Weight status Vs Average Charges charts are also created. The "Dashboard-Manual" was created in separate sheet and insights are marked in the same sheet.



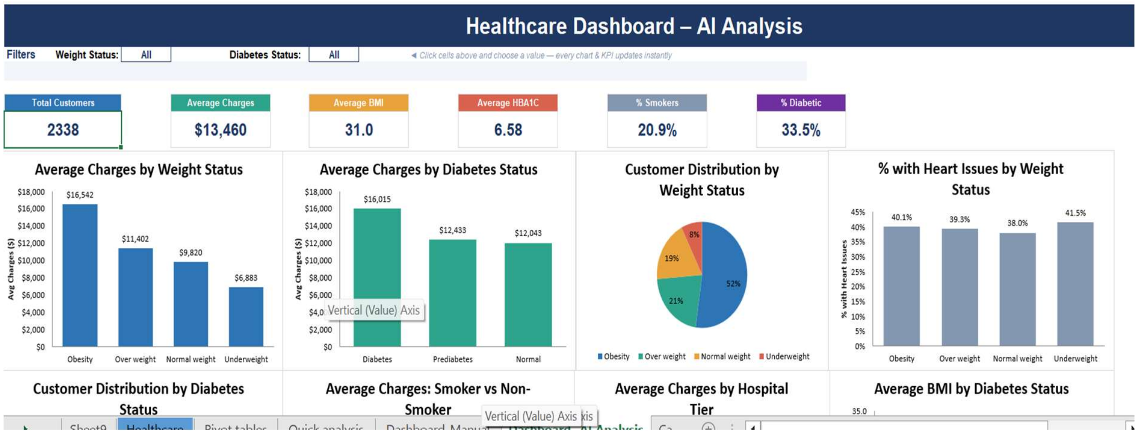
AI-Powered Summary Statistics & Visualization

The new sheet "Healthcare" was highlighted from other sheets.Using "Data Analysis" tool in the Data tab the automatically generated statistical analysis for healthcare data was created in separate sheet. When we select the healthcare column, the "Quick Analysis" tool was apeared in the sheet. Using the **"Quick analysis"** total customers count, total hospitalisation count, total healthcare costs, average costs and medical metrics are calculated and marked in separate sheet **Quick Analysis.** The recommended charts also added in this sheet.



AI-Dashboard Creation

Using **"Claude AI"** the "Healthcare Dashboard – AI Analysis" was created in separate sheet.


Healthcare Dashboard – AI Analysis
Filters Weight Status: All Diabetes Status: All
◄ Click cells above and choose a value — every chart & KPI updates instantly
Total Customers 2338
Average Charges $13,460
Average BMI 31.0
Average HBA1C 6.58
% Smokers 20.9%
% Diabetic 33.5%
Average Charges by Weight Status
$16,542 $11,402 $9,820 $6,883
Obesity Over weight Normal weight Underweight
Average Charges by Diabetes Status
$16,015 $12,433 $12,043
Diabetes Prediabetes Normal
Customer Distribution by Weight Status
52% 21% 19% 8%
Obesity Over weight Normal weight Underweight
% with Heart Issues by Weight Status
40.1% 39.3% 38.0% 41.5%
Obesity Over weight Normal weight Underweight
Customer Distribution by Diabetes Status
Average Charges: Smoker vs Non-Smoker
Average Charges by Hospital Tier
Average BMI by Diabetes Status